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# SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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## ABSTRACT

Fingerprinting Large Language Models (LLMs) provides a general mechanism for provenance verification and model attribution across model-development pipelines. Most existing fingerprinting methods are evaluated after fine-tuning or deployment-time adaptation, when models have already accumulated comparatively stable signatures from training data, optimization trajectories, alignment procedures, or hyperparameter choices. Because much of a model’s capacity and knowledge is formed during pretraining rather than downstream fine-tuning, lineage verification during large-scale pretraining is a broader and more foundational setting for model identity. We show that existing fingerprinting methods become *unreliable* in this setting, since their post-hoc signatures typically emerge only after substantial training. This limitation departs from the classical Galton view of a fingerprint as an intrinsic identity marker that is present early and persists across later conditions. We therefore propose a more broadly applicable intrinsic notion of LLM fingerprinting: **SeedPrints**, which uses random-initialization biases as persistent, seed-dependent identifiers that exist before training begins. We show that untrained models already exhibit reproducible prediction biases determined by the initialization seed, and that these weak signals remain statistically detectable through subsequent training, enabling high-confidence lineage verification. Unlike prior techniques whose evidence weakens during early pretraining or under distribution changes, **SeedPrints** remains effective across the training stages studied here, from initialization through large-scale pretraining and downstream adaptation. Experiments on LLaMA-style and Qwen-style models demonstrate seed-level distinguishability and support birth-to-lifecycle identity verification. Evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks further show robustness under prolonged training, domain shifts, and parameter modifications.

Together, these results indicate that initialization can imprint a unique and persistent identity on LLMs, offering a “Galtonian” fingerprint within the model families and deployment transformations evaluated in this work. Code will be released upon acceptance.

## 1 INTRODUCTION

LLM fingerprints are increasingly studied as a way to identify, attribute, and trace language models from their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). In a broad provenance setting, such methods are meant to give model owners a verifiable connection between a queried or suspicious model and a candidate source model, thereby supporting audits for model theft, unauthorized reuse, or unreported derivative training (Yoon et al., 2025; Zhang, 2025).

Much of this literature invokes the biological fingerprint metaphor introduced by Francis Galton’s *Finger Prints* (1892) (Galton, 1892):

## SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

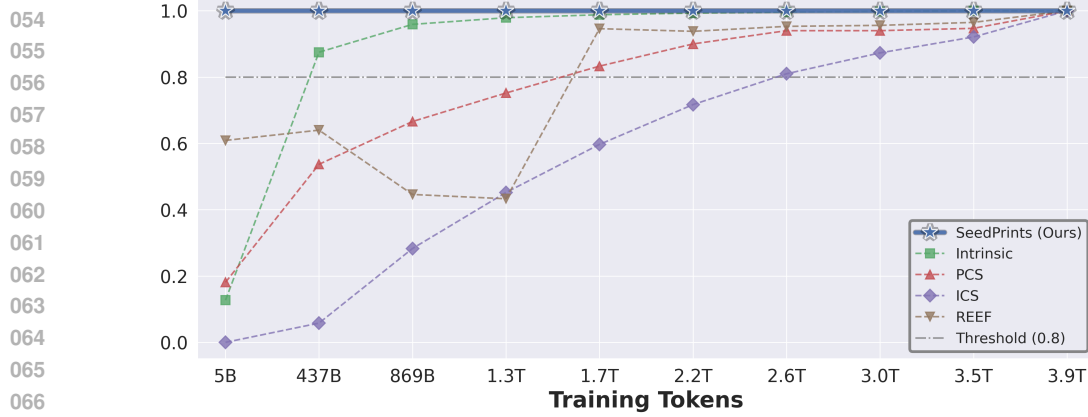


Figure 1: **Existing fingerprinting methods do not reliably detect lineage during early pre-training.** We compare five methods on OLMo-2-7B checkpoints ranging from 5B to 3.9T training tokens, with each checkpoint tested against the final checkpoint. The y-axis reports the similarity score, where larger values indicate stronger lineage evidence; the dashed line marks the 0.8 detection threshold. Although all baselines weaken and fall below the threshold for early checkpoints, **SeedPrints** detects lineage perfectly ( $p \ll 0.001$ , plotted as  $1 - p$ ) from the first available checkpoint onward.

*“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”*

Taken seriously, this analogy asks for fingerprints that are unique, persistent, and present from a model’s “birth” at initialization, rather than only after a particular training or adaptation history. Existing fingerprinting approaches often do not meet this stronger standard (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As illustrated in Figure 1, they can fail to verify lineage at early pretraining checkpoints. The reason is that many signatures are defined after a model has largely converged, for example by extracting patterns from learned parameters or generated text. Their separability can therefore reflect training data, optimization dynamics, hyperparameters, or alignment artifacts rather than a birthmark of the model itself. In Section 5.1, for instance, existing baselines often misidentify lineage when the training data distribution changes substantially. These methods are consequently better understood as post-hoc identifiers, whereas a **Galtonian fingerprint** should be an innate and persistent mark that follows a model throughout development.

In this work, we study a stricter and more generally useful notion of LLM fingerprinting: an intrinsic property that is already present at *initialization* and remains detectable at *any* later point in the training process. Our key contributions are:

- **Identifying a broad limitation of existing methods.** We show that prior fingerprinting approaches do not reliably verify lineage during early pretraining and can be misled by substantial shifts in training data distribution. (Section 5.1)
- **Discovering and leveraging initialization-driven fingerprints.** We find that untrained models exhibit seed-dependent biases in their internal representations, yielding a weak but persistent identity signal from initialization. (Section 3) We then introduce **SeedPrints**, a method that isolates these initialization-born fingerprints and is less dependent on confounding factors such as data distribution, training duration, and optimization dynamics. (Section 4)
- **Empirical validation across model families and deployment conditions.** We show that our method distinguishes models that differ only in initialization seed, even when they share the same training pipeline and data order, and that the signal persists under continual training across multiple datasets. (Sections 5.1 and 5.2) We further evaluate on *LeaFBench* (Shao et al., 2025), covering practical post-development transformations including continued pretraining, instruction tuning, PEFT, quantization, merging, and distillation. (Section 5.3)

## 2 RELATED WORK

LLM protection methods can be viewed as two broad families: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery

et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* signatures from pre-existing model behaviors without modifying the model (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025). This distinction matters for generalization because active methods assume control over a training or serving process, whereas passive methods can in principle support retrospective audits of already-released or third-party models.

**Watermarking and Active Fingerprinting.** Active approaches intentionally implant verifiable identifiers that can later support ownership checks. Classical techniques include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures, and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, common instantiations include *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.), and *model weight watermarks*, which embed identifiers into parameters or associate them with secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints can be simple to verify and may survive moderate fine-tuning (e.g., Dasgupta et al. 2024), these invasive schemes require prior control of the model-development process, limiting their applicability for retroactive provenance analysis of independently trained or third-party models.

**Passive LLM Fingerprinting.** Passive fingerprinting seeks to identify models from intrinsic properties or observable behavior without injecting a new marker. The available signals depend on access assumptions. With **white-box access**, signatures can be extracted from weights or internal representations, using properties such as attention-matrix distributions (Yoon et al., 2025), kernel alignment of hidden representations (Zhang et al., 2024), or stable directions in parameter space. In **black-box settings**, methods instead analyze input-output behavior through crafted queries (Pasquini et al., 2024), distinctive prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025), or iterative prompting-response games (Iourovitski et al., 2024), although such signals can be more sensitive to fine-tuning and serving policies. These approaches typically define signatures *post hoc*: they identify properties that emerge from a completed training, alignment, or deployment process rather than fingerprints present at random initialization. Consequently, they are not designed for lineage verification throughout large-scale pretraining, where many post-training signals have not yet formed. Our work targets this broader all-stage setting by asking whether initialization itself can provide the “Galtonian” identity signal.

### 3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

This section establishes the empirical observation behind our method: language models can exhibit prediction biases that originate from the random initialization seed, and those biases remain detectable after training rather than being fully overwritten by data or optimization.

**Initialization induces seed-specific bias profiles.** We evaluate a LLaMA-2-style model initialized with seed 123 on 10,000 random input sequences of length 1,024, with tokens sampled uniformly from the vocabulary. In Figure 2 (Left), we report how often each output token (upper panel, final logits) and each hidden dimension (lower panel, last-layer representations) attains the minimum value across the 10,000 random trials. The strongly non-uniform profiles show that even untrained models can have extreme output-preference biases rather than uniform responses. When the same protocol is repeated for other initialization seeds, the overall bias magnitude remains comparable, but the particular argmin dimensions vary by seed (Figure 2, Lower Right). These observations align with recent findings that biases in randomly initialized models can arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and amplified by self-attention (Li et al., 2026). From this perspective, different initialization seeds select different contraction directions, producing distinct argmin patterns across output dimensions.

**Training preserves initialization-born bias** Training substantially changes output magnitudes, but it does not entirely erase the relative preferences over output dimensions that are induced at initialization. We train the same initialized model (seed 123) for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate intermediate checkpoints. We focus on the  $m = 50$  output dimensions most frequently assigned minimum values by the initialized model. For each checkpoint, we compute the correlation between its responses and the initialized model’s responses on random inputs, restricted to these dimensions. This statistic measures whether a later checkpoint continues to rank or disfavor the same dimensions in a similar input-wise manner. As an unrelated-lineage baseline, we compute

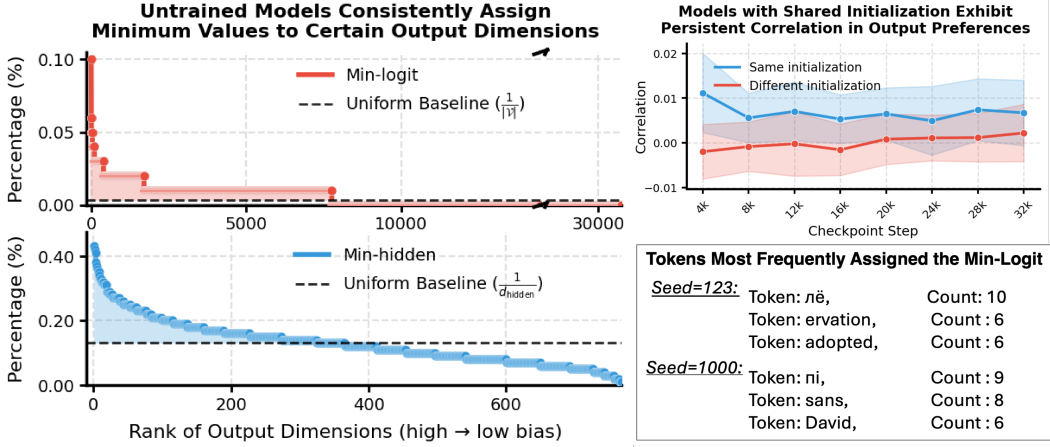


Figure 2: **Initialization-born output bias persists through training.** *Left:* Under completely random inputs, a randomly initialized LLaMA-2-style model produces outputs that are far from uniform: some dimensions are systematically disfavored and frequently receive the minimum value across random inputs. This extreme bias appears in both logits (top, red) and final hidden representations (bottom, blue). The dashed line gives the expected frequency under a uniform distribution. The arrows in the top panel mark a broken x-axis that omits low-frequency tail ranks. *Upper Right:* As training proceeds, models retain weak input-wise correlations in output bias with the base model from which they are trained. The same-lineage correlation distribution is consistently shifted upward relative to correlations with independently initialized base models, showing that initialization-born bias remains measurable after training. *Lower Right:* The dimensions with strong bias are seed-dependent; the example shows tokens that most often receive the minimum logit.

the same correlation against independently initialized models (seed 1000). As shown in Figure 2 (Upper Right), the absolute correlations are small, but same-initialization correlations are consistently shifted upward relative to the independent baseline and stabilize rather than decaying to zero. This suggests that initialization leaves a weak yet statistically detectable bias signal that can persist across the training process.

#### 4 ALGORITHM

Section 3 shows that models sharing an initialization lineage can retain weak but consistent agreement on a subset of output dimensions, such as dimensions disfavored by the untrained model under random inputs. The algorithmic goal is therefore broader than matching a known checkpoint: given any two models, without access to the true initialized model, we identify dimensions that may carry shared bias signals and test whether the observed agreement is statistically significant. The full procedure is summarized in Algorithm 1, with each component described below.

**Extract identity dimensions between any two models.** Let  $X = \{x_i\}_{i=1}^n$ , where each  $x_i \in \mathbb{R}^{\ell \times d}$  is a random sequence of length  $\ell$ . Random inputs can be instantiated either by uniformly sampling  $\ell$  token IDs from the vocabulary or by directly sampling  $\ell$  independent vectors from a  $d$ -dimensional isotropic Gaussian distribution. For any model  $g$ , we define the mean response vector  $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$ , where  $d_{\text{out}}$  is the output dimensionality and  $g(x_i)$  may denote either logits or final-layer representations. We identify the model’s disfavored dimensions as the bottom- $m$  coordinates of  $\bar{g}$ :

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We use the mean response vector to extract bias signals because it gives a stable, noise-robust alternative to per-sample argmin frequency, i.e., counting how often each dimension attains the minimum value as in Figure 2. This choice also performs well empirically, and Appendix C.1 provides a formal comparison.

Given two models  $f$  and  $f'$ , we define their *identity dimensions* as the intersection  $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$ . If two models share an initialization lineage, they should independently recover a similar set of disfavored dimensions, producing non-trivial overlap between their bottom- $m$  sets. The intersection criterion therefore provides mutual verification: it reduces spurious agreement caused by architecture

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**Algorithm 1** Distribution Correlation Test on Identity Dimensions

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**Require:** base model  $f$ , suspicious model  $f'$ ; random inputs  $X = \{x_i\}_{i=1}^n$ ; fingerprint size  $m$ ; significance level  $\alpha$

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs  $\bar{f}, \bar{f}'$  over  $X$
- 2:  $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$  bottom- $m$  dimensions of  $\bar{f}, \bar{f}'$
- 3:  $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$  (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize  $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$  for each  $x_i \in X$  across identity dimension
- 5: **for** each  $s_j \in \mathcal{S}$  **do**
- 6:    $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$  (correlation in in input-wise preferences)
- 7: **end for**
- 8:  $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{T}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct  $\mathcal{T}_{\text{null}}$  by applying the same pipeline to two Gaussian matrices  $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test  $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$  vs.  $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage  $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

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or common training dynamics, while retaining bias signals more plausibly tied to shared initialization. When one model is closer to initialization, its disfavored dimensions can also serve directly as an anchor.

**Correlation statistics.** After restricting both models to the identity dimensions  $\mathcal{S}$ , we compare their responses over the shared random inputs.<sup>1</sup> For each  $j \in \mathcal{S}$ , we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n).$$

This produces a set of correlation statistics  $\mathcal{T} = \{\tau_j\}_{j \in \mathcal{S}}$  that summarizes agreement between the two models across identity dimensions. As observed in Section 3, same-lineage models tend to exhibit a distributional upward shift in these correlations compared with unrelated models.

**Hypothesis testing via null distribution.** Because same-lineage and different-lineage correlation distributions can overlap, a fixed threshold on individual statistics is not sufficient for broadly reliable decisions. We instead test whether the observed correlations are significantly larger than expected under the null hypothesis of no lineage relation. The most direct null would use correlations between many independently initialized models, as in Figure 2, but constructing that empirical baseline is computationally expensive. Appendix B shows that the null correlation distribution between independently initialized models is well approximated by a Gaussian distribution. This motivates a cheaper simulation-based alternative: constructing an empirical null distribution from Gaussian surrogates passed through the same pipeline, as summarized in Algorithm 1.

Although this simulation-based null is already a useful empirical approximation, it adds avoidable randomness and computational cost: its precision depends on the number of simulation trials, results can vary across runs, and comparing two empirical distributions requires assumptions from the chosen test, such as the one-sided t-test or the Mann–Whitney U test. Since the Kendall–Tau null distribution under weak independence has a closed form, we use an analytical test as the main procedure while also providing empirical and analytical implementations in the code repository. Under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance  $\sigma^2$ . By the central limit theorem, as the number of identity dimensions  $|\mathcal{S}|$  grows, the average statistic  $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$  is well approximated by  $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$ . Softmax normalization introduces weak dependencies across dimensions, but the relatively high temperature ( $T = 10$ ) reduces these effects, making the independence approximation reasonable in practice. Appendix B empirically validates that this Gaussian approximation closely matches the distribution produced by the full pipeline. We compute the one-sided z-score  $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$  and the corresponding p-value  $p = 1 - \Phi(z)$ , where  $\Phi$  is the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage if  $p < 0.01$ .

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<sup>1</sup>Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Table 1: Seed-level fingerprint comparison between models initialized with different seeds. Table 2: Trained descendants retain the fingerprint behavior of their initialization ( $p$ -value  $< 0.01$ ).

| Model Pairs               | $p$ -value ( $> 0.01$ ) | Model Pairs                             | SeedPrints | Baselines |        |        |        |
|---------------------------|-------------------------|-----------------------------------------|------------|-----------|--------|--------|--------|
|                           |                         |                                         | $p$ -value | Intrinsic | REEF   | PCS    | ICS    |
| $s_{42}$ vs. $s_{2000}$   | 0.116                   | $s_{42}^{init}$ vs. $s_{42}^{base}$     | 3.37e-26✓  | -0.021×   | 0.375× | 0.580× | 0.196× |
| $s_{123}$ vs. $s_{42}$    | 1.000                   | $s_{123}^{init}$ vs. $s_{123}^{base}$   | 3.16e-33✓  | 0.149×    | 0.369× | 0.581× | 0.188× |
| $s_{1000}$ vs. $s_{123}$  | 0.327                   | $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 1.12e-21✓  | -0.252×   | 0.381× | 0.581× | 0.188× |
| $s_{2000}$ vs. $s_{1000}$ | 1.000                   | $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 3.70e-31✓  | -0.337×   | 0.331× | 0.581× | 0.188× |

**Practical instantiation.** The method can use either logits or hidden representations as model outputs. Both choices fit the formulation, but logits require many more random inputs as vocabulary size grows, which increases computational overhead (see Appendix C.2 for theoretical analysis). For efficiency and stability across the evaluated open-weight models, we use final-layer hidden states as the default output. The resulting  $p$ -value gives a direct statistical lineage test, avoiding heuristic similarity thresholds.

## 5 EXPERIMENTS

This section evaluates whether initialization-level fingerprints generalize beyond a single checkpoint or training condition. In Section 5.1, we test two properties expected of a genuine fingerprint: (i) it enables *birth verification at the seed level*, and (ii) it remains *verifiable across the entire training lifecycle*. Existing baselines do not support early-pretraining verification and often rely on data-dependent signals that break under substantial distribution shifts. We report main results using the analytical null with *hidden state* outputs, and provide additional empirical-null results with both *logit* and *hidden state* outputs, evaluated by the one-sided  $t$ -test and Mann–Whitney  $U$  test, in Appendix D.2. In Section 5.3, we evaluate all methods in practical infringement scenarios using *LeaFBench* (Shao et al., 2025). Across 65 model instances and 6 representative post-development techniques, our method is competitive with the strongest post-training baselines while preserving robustness across diverse deployment transformations.

**Baselines** We compare against four passive fingerprinting baselines that cover weight- and representation-based attribution signals. *Intrinsic fingerprint* (Yoon et al., 2025) (also referred to as *PDF* in some papers) compares models using the similarity of layerwise standard-deviation profiles of attention parameters. *REEF* (Zhang et al., 2024) computes centered-kernel-alignment (CKA) similarity between feature representations produced by two models on the same samples. *PCS* and *ICS* (Zeng et al., 2024) (collectively called *HuRef* in some papers) are weight-similarity methods: PCS flattens all parameters and measures cosine similarity, whereas ICS constructs invariant terms from the weights and compares those invariants by cosine similarity. In Section 5.3, we also include the weaker gradient-based fingerprint *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), binary baseline decisions use a **0.8 similarity threshold**.

Across the experiment tables, cell colors encode lineage relationships: **green** marks models from the same source, whereas **red** marks models from different sources. For example,  $s_{42}^{init}$  vs.  $s_{42}^{base}$  compares a seed-42 initialization with its continued-pretrained descendant, so it is green. By contrast,  $s_{2000}^{init}$  vs.  $s_{42}^{base}$  compares a seed-2000 initialization with a model trained from seed 42, so it is red. We use ✓ for correct detections and × for errors.

### 5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main text reports LLaMA-style results, while Appendix D reports the corresponding Qwen-style results. The two model styles yield consistent conclusions.

**Different initialization seeds produce distinct fingerprints** Table 1 reports  $p$ -values from correlation tests between model pairs initialized with different random seeds (42, 123, 1000, and 2000). All  $p$ -values are consistently  $> 0.01$ , so the method separates models whose only relevant distinction is the seed. This supports seed-level identity: different random initializations produce distinguishable fingerprint behaviors already “at birth.”

Table 3: Shared data and training order do not make fingerprint behavior converge across different initializations.

| Model Pair                              | $p$ -value ( $> 0.01$ ) |
|-----------------------------------------|-------------------------|
| $s_{42}^{init}$ vs. $s_{123}^{base}$    | 0.573                   |
| $s_{123}^{init}$ vs. $s_{2000}^{base}$  | 0.724                   |
| $s_{2000}^{init}$ vs. $s_{1000}^{base}$ | 0.109                   |
| $s_{1000}^{init}$ vs. $s_{42}^{base}$   | 0.883                   |

Table 4: Fingerprint persistence under continued training across different data domains (base model: seed 1000, corpus openwebtext).

| Setting                 | SeedPrints    | Baselines |        |        |        |
|-------------------------|---------------|-----------|--------|--------|--------|
| Continual corpus (seed) | $p$ -value    | Intrinsic | REEF   | PCS    | ICS    |
| TinyStories (1000)      | $\approx 0$ ✓ | 1.000✓    | 0.759× | 0.999✓ | 0.996✓ |
| TinyStories (123)       | 1.000✓        | 0.950×    | 0.658✓ | 0.332✓ | 0.012✓ |
| the_stack (1000)        | $\approx 0$ ✓ | 0.489×    | 0.557× | 0.585× | 0.123× |
| the_stack (123)         | 1.000✓        | 0.445✓    | 0.580✓ | 0.301✓ | 0.026✓ |

**Training preserves the initialization fingerprint.** Table 2 compares each initialization model  $s_i^{init}$  with its descendant  $s_i^{base}$  after training on OpenWebText (Gokaslan et al., 2019) ( $\approx 10$ B tokens). For every seed–model pair, the  $p$ -values are  $< 0.01$ , showing that the bias profiles remain strongly correlated and therefore indicate shared lineage. Thus, within this training setup, the trained model inherits the fingerprint of its initialization. The baseline methods fail to distinguish across seeds, suggesting that their separability is driven by training-induced artifacts rather than initialization-level identity.

**Identical data and order do not make fingerprints converge** In Table 3, all four “suspicious” models  $s_i^{base}$  for  $i \in \{42, 123, 100, 2000\}$  are trained on *exactly the same corpus* (OpenWebText) and in the same data order by fixing the training seed; their initialization seeds  $i$  are the only difference. Across all cross-seed pairs,  $p$ -values remain  $> 0.01$ , contrasting with the near-zero values in Table 2. Thus, shared data and curriculum do not collapse the fingerprints: the identity signal remains seed-specific.

**Continual training on diverse datasets does not confound the fingerprint** The preceding experiments establish that SeedPrints behaves like a biometric fingerprint by separating initialization identities and tracking them through training. For copyright auditing, however, distinguishing seeds is not by itself the central requirement because initialization seeds have no clear copyright status. The more general risk is that attribution methods can be misled by *data distribution, failing to recognize lineage when the training distribution shifts substantially during continued training*.

In Table 4, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two distinct datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base with (ii) *distractors* derived from a different base model (seed 123). Because the baselines are not sensitive to initialization differences, the distractor base is pretrained with a different OpenWebText data order before being continually trained on the *same* target corpus. This setup asks whether attribution methods recover the true shared lineage rather than merely matching the downstream data domain.

All prior baselines fail in the code setting (The Stack), misclassifying true descendants as distractors. This behavior indicates that their signals largely track domain similarity rather than lineage identity: TinyStories is closer to OpenWebText, whereas The Stack differs sharply, and the larger distribution shift bypasses detection. SeedPrints correctly attributes lineage for both corpora. The fingerprint is therefore not merely a proxy for data distribution; in these experiments, it survives substantial domain shift and remains detectable beyond initial pretraining.

## 5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

**Controlled pretraining trajectory.** We first ask whether the method can identify descendants throughout a controlled training trajectory. Using OpenWebText with a known initialization seed, we test lineage at intermediate checkpoints and observe consistent detection across all evaluated stages (Appendix D.2.1, Figure 5).

**Large-scale pretraining.** To test lineage in a more realistic foundation-model regime, we analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B  $\rightarrow$  3.9T tokens), treating the final checkpoint as the target and earlier checkpoints as candidate ancestors. Unlike the controlled OpenWebText experiment, this trajectory involves heterogeneous corpora, longer optimization, and

Table 5: Lineage-verification results under large-scale fine-tuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the  $p$ -value from our correlation test ( $< 0.01$  indicates a strong signal). Four baselines all report similarity scores (threshold = 0.8, higher = better).

| Model                               | # Tokens | SeedPrints $p$   | Intrinsic $\uparrow$ | REEF ( $\uparrow$ ) | PCS ( $\uparrow$ ) | ICS ( $\uparrow$ ) |
|-------------------------------------|----------|------------------|----------------------|---------------------|--------------------|--------------------|
| Llama-2-finance-7B (Heenan, 2023)   | 5M       | $10^{-41955}$ ✓  | 1.0000✓              | 0.9950✓             | 0.9979✓            | 0.9952✓            |
| Vicuna-1.5-7B (Chiang et al., 2023) | 370M     | $10^{-103043}$ ✓ | 1.0000✓              | 0.9985✓             | 0.9985✓            | 0.9949✓            |
| Wizardmath-7B (Luo et al., 2023)    | 1.8B     | $10^{-180550}$ ✓ | 1.0000✓              | 0.9979✓             | 1.0000✓            | 0.9994✓            |
| Meditron-7B (Chen et al., 2023)     | 48B      | $10^{-42566}$ ✓  | 0.9990✓              | 0.9978✓             | 1.0000✓            | 0.9817✓            |
| CodeLlama-7B (Meta AI, 2024)        | 500B     | $10^{-3552}$ ✓   | 0.9480✓              | 0.9947✓             | 0.6863✗            | 0.3369✗            |
| Llemma-7B (Azerbayev et al., 2023)  | 700B     | $10^{-5136}$ ✓   | 0.9470✗              | 0.9984✓             | 0.6682✗            | 0.2905✗            |

substantial representation drift. We plot  $1 - p$  so that larger values align with similarity-based baselines. All  $p$ -values are  $\ll 0.01$  and numerically close to zero.

In Figure 1, **all existing baselines fail to verify lineage during early pretraining**: within the first **trillion** tokens,  $ICS \approx 0$ ,  $PCS < 0.3$ , and  $REEF < 0.6$ , while SeedPrints remains verifiable and becomes stronger over training. This exposes a broad evaluation gap: **many prior studies assess lineage only after fine-tuning, where detection is easier**. Our results show that large-scale pretraining can substantially reshape representations and amplify some lineage signals, which may create a *false sense of safety* if only late-stage checkpoints are tested. Lineage verification should therefore include early pretraining stages, where misuse may be hardest to detect.

**Large-scale finetuning.** We also compare our method with existing baselines under standard fine-tuning evaluations. Specifically, we test suspect models fine-tuned from Llama-2-7b (base model) using data volumes from 5 million to 700 billion tokens. The suspects cover diverse downstream variants, including Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), Code-Llama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023). Their fine-tuning data volumes are 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As shown in Table 5, SeedPrints maintains  $p < 0.01$  across all settings.

### 5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world model releases often apply *parameter-altering* adaptations to foundation models, including fine-tuning, PEFT, and quantization, and these transformations can weaken or distort fingerprint signals. To evaluate whether SeedPrints remains useful under such deployment conditions, we use *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing.

**Across Source Models.** We audit two source-model types: *Pre-Trained (PT)* models and *Instruction-Tuned (IT)* models. For each source type, the task is to separate *derivative* models produced by post-training adaptations from *independent* models trained without the source weights. This mirrors common provenance audits in which the claimed source may be either a PT base or an IT variant. The evaluation includes **65** models in total, with model names and HuggingFace repositories listed in Appendix E.

**Metrics.** We report detection performance with (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), which measures the maximum separation between derivative-model and independent-model score distributions.

SeedPrints produces per-pair  $p$ -values from statistical tests, whereas baselines output similarity scores. These quantities have different semantics: similarity scores are typically treated as linear measures, while  $p$ -values are tail probabilities describing how rare an observation is under the null hypothesis and should not be interpreted linearly. To compare with baselines that compute AUC from similarity scores, we convert  $p$ -values to scores via  $s = 1 - p$ . This conversion is convenient but not ideal for SeedPrints, because very small  $p$ -values vary on an exponential scale while threshold sweeps are linear, making fine-grained differences difficult to represent. Even under this comparison protocol, our method remains competitive. For evaluation, we (i) compute AUC for all methods by sweeping thresholds over scores  $s$ , and (ii) report KS as a threshold-free measure of distributional separability. This keeps the protocol compatible with baseline pipelines while retaining the statistical interpretation of the SeedPrints decision rule. We omit Manhattan distance because absolute distances between  $p$ -values are not meaningful.

Table 6: Performance comparison of LLM fingerprinting methods across source-model families. “PT” and “IT” denote pre-trained and instruction-tuned source models, respectively. Per-family results are shown for the 6 families containing both PT and IT sources. The “Overall” scores are computed over all test model pairs rather than averaged across families.

| Method     | Metric                  | Qwen2.5-7B |       | Qwen2.5-14B |       | Llama3.1-8B |       | Mistral-7B |       | Gemma2-2B |       | Llama2-7B |       | Overall      |
|------------|-------------------------|------------|-------|-------------|-------|-------------|-------|------------|-------|-----------|-------|-----------|-------|--------------|
|            |                         | PT         | IT    | PT          | IT    | PT          | IT    | PT         | IT    | PT        | IT    | PT        | IT    |              |
| REEF       | AUC $\uparrow$          | 0.796      | 0.862 | 0.947       | 0.913 | 1.000       | 0.999 | 0.997      | 0.997 | 0.963     | 0.968 | 0.750     | 0.706 | <b>0.915</b> |
|            | KS Statistic $\uparrow$ | 0.492      | 0.633 | 0.875       | 0.735 | 0.987       | 0.987 | 0.975      | 0.980 | 0.950     | 0.837 | 0.750     | 0.694 | <b>0.739</b> |
| Gradient   | AUC $\uparrow$          | 0.657      | 0.743 | 0.763       | 0.789 | 0.897       | 0.904 | 0.872      | 0.682 | 1.000     | 1.000 | 0.650     | 0.768 | <b>0.801</b> |
|            | KS Statistic $\uparrow$ | 0.354      | 0.423 | 0.500       | 0.563 | 0.646       | 0.705 | 0.725      | 0.500 | 1.000     | 1.000 | 0.275     | 0.479 | <b>0.508</b> |
| ICS        | AUC $\uparrow$          | 1.000      | 1.000 | 0.997       | 0.997 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.994</b> |
|            | KS Statistic $\uparrow$ | 1.000      | 1.000 | 0.975       | 0.979 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.943</b> |
| Intrinsic  | AUC $\uparrow$          | 1.000      | 1.000 | 1.000       | 1.000 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.994</b> |
|            | KS Statistic $\uparrow$ | 1.000      | 1.000 | 1.000       | 1.000 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.989</b> |
| SeedPrints | AUC $\uparrow$          | 0.992      | 0.994 | 0.988       | 0.990 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.992</b> |
|            | KS Statistic $\uparrow$ | 0.985      | 0.987 | 0.975       | 0.980 | 1.000       | 1.000 | 1.000      | 1.000 | 1.000     | 1.000 | 1.000     | 1.000 | <b>0.986</b> |

In many practical audits, calibrating an optimal similarity threshold is infeasible. Although we report AUC for comparability, SeedPrints differs from prior methods by providing a direct statistical test: **given any pair of models, it outputs a  $p$ -value that supports a lineage decision without relying on tunable similarity thresholds.**

**Parameter-Altering Techniques.** To assess robustness across common post-development transformations, we evaluate models produced by (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

### 5.3.1 FAMILY-WISE EFFECTIVENESS

The benchmark contains seven model families. For each family, we treat the pretrained base model (PT) and instruction-tuned model (IT) as separate sources and compare each source with all other models in the 65-model pool. Table 6 reports the results. We omit TinyLlama-1.1B families in Table 6 because they are trivial for all methods. Each column corresponds to one family, with AUC computed over the relevant tests using a family-calibrated threshold; the overall score is computed across all test pairs using a global threshold, so it is not the average of per-family scores. *Intrinsic*, *ICS*, and *SeedPrints* are all nearly saturated across alterations, with overall AUC of 0.994, 0.994, and 0.992, respectively, and KS of 0.989, 0.943, and 0.986. For these methods, most family-wise scores reach the full score. The remaining baselines, *REEF* and *Gradient*, perform substantially worse. These results show that although SeedPrints is designed around intrinsic seed-level fingerprints, it also functions as a biometric-like identity tracker across the evaluated model families.

### 5.3.2 ALTERATION-WISE ROBUSTNESS

Across alteration types, *Instruct* is consistently the easiest setting: even the weakest baseline, *Gradient*, reaches AUC 0.895. *Finetune* and *Merge* are more challenging overall; for example, no method achieves perfect performance on *Finetune*. The effects of *PEFT*, *Quantization*, and *Distillation* are more heterogeneous. *REEF* drops under *Quantization* ( $0.996 \rightarrow 0.871$ ) and *Distillation* ( $0.996 \rightarrow 0.858$ ), while remaining strong on *PEFT* (0.994); conversely, *PEFT* and *Quantization* are harder for *Gradient* ( $0.895 \rightarrow 0.776/0.756$ ). The three stronger methods are robust across most transformations, with all scores  $> 0.9$ . SeedPrints decreases modestly to 0.959 on *Merge*, failing on one merge model out of seventeen. This single error suggests that interpolating weights with unrelated models can dilute the initialization signal below the detection threshold. Averaged over the six alteration types, *Intrinsic* has the highest overall AUC, followed closely by *ICS* and *SeedPrints*.

## 5.4 ABLATION STUDY ON HYPERPARAMETERS

SeedPrints has three hyperparameters: the number of random input sequences  $n$ , the length of each random sequence  $l$ , and the fingerprint size  $m$ . The first two determine the amount of random-input evidence, while  $m$  controls how many low-mean coordinates enter the lineage test. In the main experiments in Sections 5.2 and 5.3, we set  $n=2000$ ,  $l=1024$ , and  $m=0.1d_{\text{out}}$ , where  $d_{\text{out}}$  denotes hidden-state dimensionality, so the fingerprint size scales with model width. Here, we vary one

Table 7: Performance of LLM fingerprinting methods across parameter-altering deployment techniques.

| Method     | Metric                  | Instruct | Finetune | PEFT  | Quantization | Merge | Distillation | Overall Avg  |
|------------|-------------------------|----------|----------|-------|--------------|-------|--------------|--------------|
| REEF       | AUC $\uparrow$          | 0.996    | 0.872    | 0.994 | 0.871        | 0.954 | 0.858        | <b>0.924</b> |
|            | KS Statistic $\uparrow$ | 0.949    | 0.637    | 0.940 | 0.812        | 0.851 | 0.615        | <b>0.801</b> |
| Gradient   | AUC $\uparrow$          | 0.895    | 0.729    | 0.776 | 0.756        | 0.832 | 0.887        | <b>0.812</b> |
|            | KS Statistic $\uparrow$ | 0.756    | 0.441    | 0.469 | 0.586        | 0.519 | 0.679        | <b>0.575</b> |
| ICS        | AUC $\uparrow$          | 1.000    | 0.987    | 1.000 | 1.000        | 0.981 | 1.000        | <b>0.995</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.925    | 1.000 | 1.000        | 0.941 | 1.000        | <b>0.978</b> |
| Intrinsic  | AUC $\uparrow$          | 1.000    | 0.980    | 1.000 | 1.000        | 1.000 | 1.000        | <b>0.997</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.971    | 1.000 | 1.000        | 1.000 | 1.000        | <b>0.995</b> |
| SeedPrints | AUC $\uparrow$          | 1.000    | 0.995    | 0.990 | 1.000        | 0.959 | 1.000        | <b>0.991</b> |
|            | KS Statistic $\uparrow$ | 1.000    | 0.990    | 0.980 | 1.000        | 0.941 | 1.000        | <b>0.985</b> |

hyperparameter at a time while fixing the others to default values (i.e.,  $n = 2000$ ,  $m = 400$ ,  $l = 768$ ), and report empirical accuracy and false-positive rates at significance level  $\alpha = 0.01$  in Table 8.

Table 8: Hyperparameter ablation under significance level  $\alpha = 0.01$ .

Increasing  $n$  and  $l$  consistently improves identification performance. This matches the theoretical analysis in Section C.2, which shows that recovering the population-level bias requires  $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$  queries, where  $\gamma$  is the bias margin between biased coordinates and the remaining dimensions. Thus, each target output dimension needs enough samples for reliable detection, explaining the weaker performance at small  $n$ . Meanwhile,  $\gamma$  increases with  $l$  through accumulation effects, so longer sequences strengthen separation. With a fixed and potentially insufficient number of sequences, increasing  $l$  reduces the query burden and improves detection.

| Ablation on $n$ | 200    | 500    | 2000   |
|-----------------|--------|--------|--------|
| ACC             | 0.7368 | 0.7778 | 0.9375 |
| Empirical FPR   | 0      | 0      | 0      |
| Ablation on $l$ | 512    | 768    | 1024   |
| ACC             | 0.9048 | 0.9375 | 1.0000 |
| Empirical FPR   | 0.04   | 0      | 0      |
| Ablation on $m$ | 200    | 400    | 800    |
| ACC             | 0.9375 | 0.9375 | 0.875  |
| Empirical FPR   | 0.04   | 0      | 0.1463 |

The effect of  $m$  is non-monotonic. If  $m$  is too small, the selected coordinate set is unstable because there is not enough empirical evidence to form a reliable distribution. If  $m$  is too large, many irrelevant coordinates enter the test and add noise, shrinking  $\gamma$  and requiring quadratically more samples to compensate. Empirically,  $m$  in the range 200–400 works well across the evaluated model sizes, consistent with the heuristic  $m \approx 0.1d_{\text{out}}$  for typical  $d_{\text{out}} \in [3000, 5000]$ .

## 6 CONCLUSION

We introduced *SeedPrints*, an intrinsic approach to LLM fingerprinting that traces model lineage back to random initialization rather than to post-hoc training artifacts. Our results show that initialization can leave a persistent “Galtonian” fingerprint on neural language models: untrained models already display stable, seed-dependent token-preference patterns, and these biases remain statistically measurable through later training. Building on this observation, we developed a distribution-correlation test over *identity dimensions* that detects shared lineage with calibrated significance. Across LLaMA- and Qwen-style families and a broad set of evaluated deployment transformations, including continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation, SeedPrints supports “birth-to-lifecycle” verification and remains robust under domain shift. These findings position initialization-level fingerprints as a practical tool for provenance and copyright auditing within the white-box settings studied here.

**Limitations and Future Directions** This work focuses on fingerprint signals rooted in initialization-stage bias and therefore occupies a specific point in the broader design space of model attribution. The proposed method requires access to model outputs such as logits or hidden states, and may not apply when only restricted API access is available. At the same time, reliable *true fingerprints* for LLMs remain an early-stage research direction: many existing approaches, especially black-box methods, depend on signals that arise later in training or from alignment-driven artifacts, and thus do not reliably protect pretrained or early-stage models. Developing all-stage fingerprints under weaker access assumptions is therefore an important next step. See full discussions in Section A.2.

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## A LLM USAGE AND LIMITATION DISCUSSIONS

### A.1 LLM USAGE

We used AI assistants, including ChatGPT and Gemini, for manuscript-level writing and formatting support. Their role was limited to grammar and style checks, caption clarity, and other surface-level edits that do not alter the scientific scope or experimental content. For programming-related tasks, we occasionally used GitHub Copilot and Claude as coding assistants, for example for code auto-completion and debugging hints.

### A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although this work identifies seed-born, training-persistent biases that can serve as fingerprint signals, the current method remains within the white-box paradigm. It may not apply when access is limited to restricted APIs, especially if responses are deterministic, safety-filtered, or otherwise stripped of the outputs needed for our test. More broadly, *true fingerprints* for LLMs remain an early-stage research direction even with white-box access. Many existing methods rely on training-induced signatures or alignment behavior, and therefore do not reliably cover pretrained or early-stage models. By analogy to human fingerprints, which are stable across age and environment, a broadly useful LLM fingerprint should persist across continued training and deployment contexts rather than merely reflect incidental training artifacts. Extending this goal to black-box access, where a substantial gap remains relative to white-box methods, is an important but challenging direction for future work. We hope this study encourages more general and principled fingerprinting frameworks for large models.

## B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The hypothesis test in Section 4 evaluates shared lineage by comparing the distribution of correlation statistics over random inputs with a null distribution. This null represents the correlations expected from two *independent models evaluated on random inputs*.

We consider two complementary approximations to this null distribution: (i) an empirical simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. This section justifies both approximations in three steps: characterizing the true null with independently initialized models, introducing the Gaussian surrogate as a tractable proxy, and validating the analytical approximation.

### B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null distribution is the correlation statistic obtained from two independent models evaluated on random inputs. We instantiate this independence by comparing independently initialized, untrained models with different random seeds. Specifically, we estimate the empirical null from  $\sim 2500$  such pairs and compute Kendall–Tau correlations on 10,000 random inputs, following Section 3. Figure 3 shows that this true null is empirically close to a zero-centered Gaussian.

### B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null with independently initialized models is expensive because it requires many models and repeated evaluations. As a tractable alternative, we use Gaussian surrogates as proxies for model outputs under the null.

This surrogate is motivated by two general considerations. First, Gaussian approximations to neural-network outputs are common in auditing settings and are supported by empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline uses relative ordering and selection operations, such as rank correlations and bottom- $m$  selection, rather than semantic structure in the inputs. Independently initialized models also have independent parameter matrices, so their null outputs should not contain strong shared structure. Sampling independent Gaussian matrices therefore gives a simple proxy for the null behavior of independently initialized models.

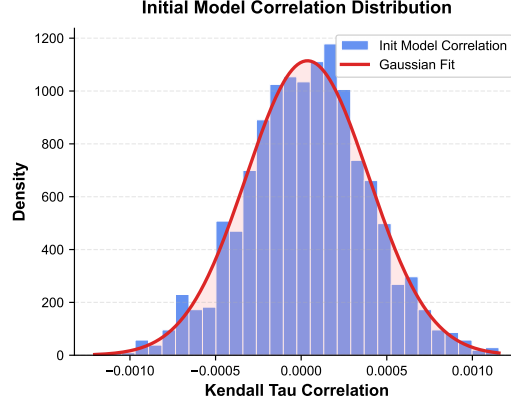


Figure 3: Empirical correlation distribution over  $\sim 2500$  pairs of independently initialized models, evaluated on random inputs, closely follows a zero-centered Gaussian distribution.

We construct the surrogate null by sampling Gaussian matrices and applying the same pipeline used for real models, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape  $\mathbb{R}^{N \times d_{\text{out}}}$ ,
2. Extract the most biased dimensions by selecting the bottom- $m$  ranked average dimensions,
3. Take the intersection set  $\mathcal{S}$  and compute correlations on it.

This construction applies the same selection procedure to Gaussian samples and therefore captures dependency patterns introduced by the pipeline itself, including those induced by selecting  $\mathcal{S}$  across dimensions. As shown in Figure 4 (middle), the resulting distribution matches the centered Gaussian shape of the empirical null from independently initialized models, supporting the Gaussian surrogate as an effective simulation-based approximation.

### B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate captures null behavior under the full pipeline, it still requires repeated sampling and introduces estimation randomness. We therefore use a more efficient analytical approximation.

Under the null hypothesis that the two models are independent<sup>2</sup>, each Kendall–Tau correlation  $\tau_j$  is approximately zero-mean with known variance  $\sigma^2$ . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces dependencies across dimensions. However, Figure 4 shows that the analytical Gaussian closely matches the Gaussian-surrogate pipeline across settings with different numbers of random inputs  $n$  and output dimensions  $D$ . This suggests that the dependencies are weak enough not to materially affect the approximation. A plausible explanation is that the relatively high temperature ( $T = 10$ ) smooths the output distribution and reduces coupling between dimensions.

Together, these results justify the analytical Gaussian null in Section 4 as an efficient, deterministic approximation that avoids costly empirical baseline construction.

**On variance scaling.** The true null from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This mismatch does not affect the hypothesis-test conclusions.

<sup>2</sup>As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

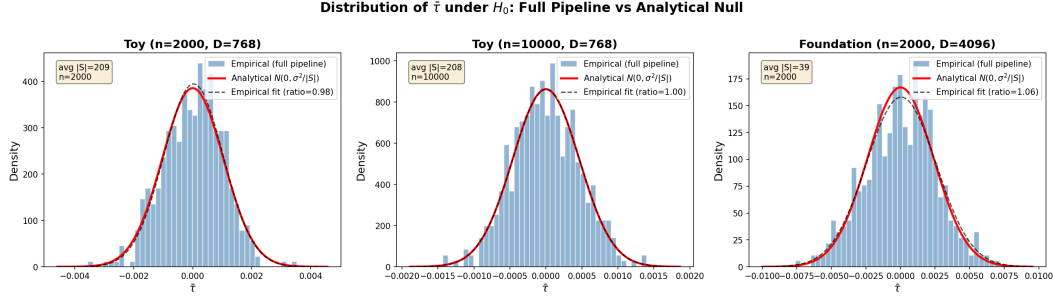


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline, including identity-dimension selection and correlation aggregation, to Gaussian surrogate outputs and compare the resulting distribution of  $\bar{\tau}$  with the analytical Gaussian  $\mathcal{N}(0, \sigma^2/|S|)$ . The close match shows that the analytical approximation captures null behavior despite weak pipeline-induced dependencies.

The primary simulation-based-null evaluation in Section D.2 uses the Mann–Whitney U-test, which is rank-based and invariant to positive scaling: for any  $c > 0$ ,  $x_i < x_j \Leftrightarrow cx_i < cx_j$ . Thus, rank statistics and rejection thresholds are unchanged. For tests that do depend on absolute variance, we have additionally evaluated variance-sensitive statistics such as the  $t$ -test, whose statistic scales as  $t' = t/c$  when variance decreases. Since  $c < 1$  in our case, scaling reduces the magnitude of  $t$  and makes the test more conservative rather than inflating significance. Empirically, across extensive experiments in Section D.2, we find that the variance scaling does not affect the power of the  $t$ -test, indicating that the distributional separation between the alternative hypothesis and the null is substantially larger than the scaled variance.

For the analytical null used in the main paper, variance appears explicitly in the  $z$ -score normalization, so scale discrepancies are absorbed in the denominator. Because the analytical variance is slightly larger than the empirical one, the resulting  $z$ -scores are smaller, making the test conservative rather than increasing false positives.

## C THEORETICAL PROPERTIES OF OUR METHODS

The method aims to identify output dimensions that an initialized model is least likely to favor under random inputs. These dimensions correspond to directions disfavored by initialization-induced bias rather than by a particular semantic input distribution, and they form the basis of the stable fingerprint signal studied in this paper.

**Setup.** Consider a model  $g$  with fixed parameters and random inputs  $x_i$  (uniform random tokens or random embeddings as in the main paper). For each output dimension  $j \in [d_{\text{out}}]$ ,  $g_j(x)$  is thus a random variable and its population mean is  $\bar{g}_j := \mathbb{E}[g_j(x)]$ . The population-level directions we aim to capture correspond to the  $m$  output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$  denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the  $m$ -th and  $(m+1)$ -th smallest means. This gap measures how well-separated the target set  $\mathcal{M}$  is from the remaining coordinates. Importantly, for a fixed  $m$  and input distribution,  $\Gamma(m)$  depends solely on the model’s inherent initialization bias. *However, increasing the length of random input sequences amplifies this bias via concentration, making the separation more pronounced at the population level.*

Two empirical strategies can extract these population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs  $X = \{x_i\}_{i=1}^n$  and select the  $m$  coordinates with the smallest

value. Following the notation in Section 4, where  $g$  denotes the model (either the base model  $f$  or the suspect model  $f'$ ):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

**2. Per-sample argmin voting.** For each random input  $x_i \in \mathbb{R}^{\ell \times d}$ , select the smallest-value coordinate and then take the  $m$  most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let  $p_j := \Pr(j^*(x) = j)$  denote the population probability that index  $j$  attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- $m$  set under voting.

Both approaches are intuitively reasonable. However, as we show below, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, making it inherently noise-sensitive.

## C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

### C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first show that the sample-mean-based operator, which selects the  $m$  smallest coordinates in the empirical mean vector, is stable under small perturbations to model outputs. The perturbations represents realistic variations in outputs from the *same* model, e.g., numerical noise, or evaluation under different precision modes.

Formally, we model perturbed outputs as  $g'(x) = g(x) + \eta(x)$ , where  $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$  represents stochastic deviations. We assume that for each sample  $x_i$  and coordinate  $j$ , the random variables  $\eta_j(x_i)$  are independent across samples (not necessarily across coordinates) and  $\sigma$ -sub-Gaussian with  $\mathbb{E}[\eta_j(x_i)] = 0$ .<sup>3</sup>

**Lemma C.1** (Stability of sample-mean-based extraction). *Let  $\hat{g}$  and  $\hat{g}'$  be the empirical means over  $X = \{x_i\}_{i=1}^n$ , i.e.,  $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$  and  $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$ . Let  $\widehat{\Gamma}(m)$  denote the empirical gap at rank  $m$ . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

**Interpretation.** The lemma provides a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decreases exponentially with the number of averaged samples  $n$ , due to variance reduction from  $\sigma^2$  per sample to  $\sigma^2/n$  in the empirical mean.

*Proof.* We condition on the sample  $X = \{x_i\}_{i=1}^n$  throughout the proof, so that  $\hat{g}$  and  $\widehat{\Gamma}(m)$  are fixed, and all probabilities are taken over the randomness of  $\{\eta(x_i)\}_{i=1}^n$  only.

For any coordinate  $j$ , by definition of  $\hat{g}'$ ,  $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$ . Each  $\eta_j(x_i)$  is mean-zero and  $\sigma$ -sub-Gaussian, so their average is mean-zero and  $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as  $1/n$  under averaging. Hence, for any  $\varepsilon > 0$ , by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

<sup>3</sup>This sub-Gaussian assumption is mild: bounded random variables are  $\sigma$ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case  $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$  is a special case of this assumption.

Now consider the event that *any* coordinate fluctuates by more than  $\varepsilon$ :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all  $d_{\text{out}}$  coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \right\}.$$

Equivalently, if  $\max_j |\hat{g}'_j - \hat{g}_j| < \widehat{\Gamma}(m)/2$ , then the bottom- $m$  identity set remains unchanged.

Let  $j_{(1)}, \dots, j_{(d_{\text{out}})}$  be a permutation of  $\{1, \dots, d_{\text{out}}\}$  such that  $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$ . Recall that the empirical gap at rank  $m$  is  $\widehat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$ . Denote  $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$  and  $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$ . For any  $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$  and any  $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$ ,  $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \widehat{\Gamma}(m)$ . With  $|\hat{g}'_i - \hat{g}_i| < \widehat{\Gamma}(m)/2$  and  $|\hat{g}'_\ell - \hat{g}_\ell| < \widehat{\Gamma}(m)/2$ , we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\widehat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\widehat{\Gamma}(m)}{2}) \geq \widehat{\Gamma}(m) - \widehat{\Gamma}(m) = 0.$$

Thus no coordinate in  $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$  can become smaller than any coordinate in  $\widehat{\mathcal{M}}^{\text{mean}}(g)$  under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since  $\widehat{\Gamma}(m)$  is fixed under conditioning on  $X$ , plugging  $\varepsilon = \widehat{\Gamma}(m)/2$  into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

### C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator on each input. Thus, arbitrarily small perturbations to outputs can change decisions with non-vanishing probability. We still consider the same stochastic perturbation model  $g'(x) = g(x) + \eta(x)$ .

**Lemma C.2** (Instability of per-sample argmin voting). *For a fixed input  $x$ , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

*denote the argmin and its margin. Then, for any fixed  $x$ ,*

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

*and for  $n$  i.i.d. inputs  $x_1, \dots, x_n$ ,*

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

*where  $\alpha := \mathbb{E}_x \left[ \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$ .*

**Interpretation.** Unlike mean extraction, voting does not benefit from variance reduction because each input contributes a hard decision based on noisy outputs. Increasing  $n$  therefore also increases the number of opportunities for index flips.

*Proof.* We first bound the per-input change probability  $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$ . Fix  $x$  and write  $j^* := j^*(x; g)$  for brevity. By definition of the margin,  $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$ . Let  $j^{(2)}$  be an index attaining this minimum, i.e.,  $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$ .

Under the perturbation  $g'(x) = g(x) + \eta(x)$ , we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since  $\eta_{j^{(2)}}(x)$  and  $\eta_{j^*}(x)$  are independent  $\mathcal{N}(0, \sigma^2)$  variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes,  $j^*(x; g') \neq j^*(x; g)$ , certainly occurs whenever  $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$ . Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let  $Z \sim \mathcal{N}(0, 1)$ . Since  $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$ , we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore,  $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$ .

Now consider  $n$  i.i.d. inputs  $x_1, \dots, x_n$ . Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across  $i$ , because the perturbations  $\eta(x_i)$  are independent across  $i$ . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define  $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$ , and  $\alpha := \mathbb{E}_x \left[ \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$ . Taking expectations over the randomness of  $x_i$ , we obtain  $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$ . Therefore, by independence across  $i$ ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function  $u \mapsto \log(1 - u)$  on  $u \in [0, 1)$ . Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set  $\widehat{\mathcal{M}}^{\text{vote}}$  is determined by the empirical counts of the dimensions  $\{j^*(x_i; g)\}_{i=1}^n$ . If the empirical gap between the  $m$ -th and  $(m+1)$ -th most frequent dimensions is of order  $O(1/n)$  (i.e., only a constant number of votes), then a change in  $j^*(x_i)$  for a single input  $x_i$  is sufficient to swap the ordering across this boundary and thus alter the top- $m$  set. Combining this observation with the lower bound  $1 - (1 - \alpha)^n$  shows that, in such regimes, the probability that  $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$  is bounded away from zero uniformly in  $n$ , i.e., per-sample argmin voting does not enjoy variance reduction with more samples.  $\square$

## C.2 INFLUENCE OF HYPERPARAMETERS

We next analyze how many random inputs are needed for the empirical mean estimator to recover the underlying bias. Reusing the notation from **Setup**, each output coordinate  $g_j(x)$  is a random variable under random inputs  $x \sim \mathcal{D}$ , with population mean  $\bar{g}_j := \mathbb{E}[g_j(x)]$ .  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$  represents the sorted means in non-decreasing order, and  $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$  is the separation margin. We estimate  $\bar{g}$  using  $n$  i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimension set is  $\mathcal{M}$  with size  $m$ , and the empirical estimate is  $\widehat{\mathcal{M}}$ .

**Lemma C.3** (Sufficient condition for recovering the true set). *If  $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$  and  $2\varepsilon < \gamma$ , then the empirical bottom- $m$  set equals the true set:  $\widehat{\mathcal{M}} = \mathcal{M}$ .*

*Proof.* For any  $j \in \mathcal{M}$  and  $j' \notin \mathcal{M}$ , we have  $\bar{g}_j \leq \bar{g}_{(m)}$  and  $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$ . The  $\ell_\infty$  bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- $m$  set.  $\square$

We next interpret this recovery condition in terms of hyperparameters. We use the same sub-Gaussian assumption as in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is  $\sigma$ -sub-Gaussian with  $\sigma$  proportional to its range. In our setting, each  $g_j(x)$  is a logit or final hidden activation, which is typically bounded or light-tailed in both initialized networks, due to controlled parameter initialization, and trained networks.

**Corollary C.4** (Sample complexity under sub-Gaussian coordinates). *Suppose each  $g_j(x)$  is  $\sigma$ -sub-Gaussian. Then with probability at least  $1 - \delta$ ,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then  $\widehat{\mathcal{M}} = \mathcal{M}$  with probability at least  $1 - \delta$ .

*Proof.* For fixed  $j$ , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all  $d_{\text{out}}$  coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to  $\delta$  and solve for  $\varepsilon$  to obtain, with probability at least  $1 - \delta$ ,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If  $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$  (i.e., if  $2\varepsilon < \gamma$ ), then Lemma C.3 applies and implies  $\widehat{\mathcal{M}} = \mathcal{M}$  with probability at least  $1 - \delta$ .  $\square$

Table 9: Seed-dependent fingerprint behavior across model families under different random initializations.

| (a) LLaMA-style model family |                |                |                |                | (b) Qwen-style model family |                |                |                |                |
|------------------------------|----------------|----------------|----------------|----------------|-----------------------------|----------------|----------------|----------------|----------------|
| Seed Pair                    | Logits Output  |                | Hidden State   |                | Seed Pair                   | Logits Output  |                | Hidden State   |                |
|                              | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |                             | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |
| $s_{42}$ vs. $s_{2000}$      | 0.404          | 0.456          | 0.357          | 0.532          | $s_{123}$ vs. $s_{1000}$    | 0.741          | 0.727          | 0.094          | 0.074          |
| $s_{123}$ vs. $s_{42}$       | 0.214          | 0.295          | 0.678          | 0.565          | $s_{1000}$ vs. $s_{123}$    | 0.954          | 0.971          | 0.125          | 0.094          |
| $s_{1000}$ vs. $s_{123}$     | 0.219          | 0.246          | 0.363          | 0.335          | $s_{42}$ vs. $s_{2000}$     | 0.273          | 0.360          | 0.451          | 0.529          |
| $s_{2000}$ vs. $s_{1000}$    | 0.282          | 0.291          | 0.434          | 0.481          | $s_{2000}$ vs. $s_{42}$     | 0.215          | 0.206          | 0.230          | 0.295          |

Table 10: (LLaMA-style model family) Trained descendants retain the fingerprint behavior of their initialization ( $p$ -value  $< 0.01$ ).

| Model Pair                              | Logits Output  |                | Hidden State   |                | Baselines |        |        |        |
|-----------------------------------------|----------------|----------------|----------------|----------------|-----------|--------|--------|--------|
|                                         | <i>t</i> -test | <i>u</i> -test | <i>t</i> -test | <i>u</i> -test | Intrinsic | REEF   | PCS    | ICS    |
| $s_{42}^{init}$ vs. $s_{42}^{base}$     | 3.33e-3✓       | 1.02e-3✓       | 2.20e-8✓       | 6.28e-8✓       | -0.021×   | 0.375× | 0.580× | 0.196× |
| $s_{123}^{init}$ vs. $s_{123}^{base}$   | 2.06e-3✓       | 7.33e-3✓       | 7.09e-6✓       | 1.37e-5✓       | 0.149×    | 0.369× | 0.581× | 0.188× |
| $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 2.44e-3✓       | 4.14e-3✓       | 5.58e-4✓       | 2.81e-3✓       | -0.252×   | 0.381× | 0.581× | 0.188× |
| $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 5.63e-3✓       | 6.76e-3✓       | 4.00e-10✓      | 1.27e-9✓       | -0.337×   | 0.331× | 0.581× | 0.188× |

Thus, accurate recovery requires  $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$ : the number of random queries grows only logarithmically with output dimension but inversely with the squared separation margin  $\gamma$ . The margin  $\gamma$  itself has two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of  $m$ .** Different choices of  $m$  correspond to different parts of the ranked spectrum  $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$ , thus yielding different effective margins in practice.

The input length  $l$  controls the former factor, while the latter is selected empirically from observed gaps in the mean spectrum.

## D MORE EXPERIMENT DETAILS

### D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models with the Hugging Face Transformers Trainer (Wolf et al., 2020) and use Accelerate (Gugger et al., 2022) for distributed runs. Open-source models are loaded from their official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License and other models under Apache-2.0. Datasets are downloaded through the Hugging Face Datasets library, which is Apache-2.0, with dataset content governed by each dataset’s stated license.

### D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section provides complementary results for Section 5. We evaluate the empirical simulation-based null in Algorithm 1 using both *logits* and *hidden state* outputs, and report both the one-sided *t*-test (*t*-test) and Mann–Whitney *U* test (*U* test).

#### D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline is stochastic, we report  $p$ -values averaged over 10 independent trials and use a significance level of

Table 11: (Qwen-style model family) Trained descendants retain the fingerprint behavior of their initialization models ( $p$ -value  $< 0.01$ ).

| Model Pair                              | Logits Output |           | Hidden State |           |
|-----------------------------------------|---------------|-----------|--------------|-----------|
|                                         | $t$ -test     | $U$ -test | $t$ -test    | $U$ -test |
| $s_{123}^{init}$ vs. $s_{123}^{base}$   | 2.38e-03      | 1.68e-03  | 7.36e-15     | 3.38e-13  |
| $s_{1000}^{init}$ vs. $s_{1000}^{base}$ | 1.35e-04      | 4.84e-05  | 4.41e-13     | 2.01e-11  |
| $s_{42}^{init}$ vs. $s_{42}^{base}$     | 1.39e-03      | 1.46e-03  | 1.06e-24     | 2.05e-19  |
| $s_{2000}^{init}$ vs. $s_{2000}^{base}$ | 1.80e-03      | 1.28e-03  | 4.87e-24     | 1.92e-20  |

Table 12: Fingerprint persistence under continued training across different data domains (base model: seed 1000, corpus openwebtext).  $U$ -test refers to the Mann–Whitney  $U$  test.

(a) LLaMA-style model family

| Setting                 | Ours (logits) |            | Ours (hidden) |           | Baselines |        |        |        |
|-------------------------|---------------|------------|---------------|-----------|-----------|--------|--------|--------|
| Continual corpus (seed) | $t$ -test     | $U$ -test  | $t$ -test     | $U$ -test | Intrinsic | REEF   | PCS    | ICS    |
| TinyStories (1000)      | 0✓            | 0✓         | 0✓            | 7.77e-89✓ | 1.000✓    | 0.759× | 0.999✓ | 0.996✓ |
| TinyStories (123)       | 1.000✓        | 1.00✓      | 0.943✓        | 0.902✓    | 0.950×    | 0.658✓ | 0.332✓ | 0.012✓ |
| the_stack (1000)        | 0✓            | 1.73e-287✓ | 0✓            | 3.09e-69✓ | 0.489×    | 0.557× | 0.585× | 0.123× |
| the_stack (123)         | 0.616✓        | 0.479✓     | 0.732✓        | 0.831✓    | 0.445✓    | 0.580✓ | 0.301✓ | 0.026✓ |

(b) Qwen-style model family

| Setting                 | Ours (logits) |            | Ours (hidden) |           | Baselines |        |        |        |
|-------------------------|---------------|------------|---------------|-----------|-----------|--------|--------|--------|
| Continual corpus (seed) | $t$ -test     | $U$ -test  | $t$ -test     | $U$ -test | Intrinsic | REEF   | PCS    | ICS    |
| TinyStories (1000)      | 5.36e-09✓     | 1.92e-07✓  | 8.49e-214✓    | 5.09e-71✓ | 1.000✓    | 0.957✓ | 0.999✓ | 0.996✓ |
| TinyStories (123)       | 0.434✓        | 0.433✓     | 0.256✓        | 0.065✓    | 0.913×    | 0.199✓ | 0.328✓ | 0.039✓ |
| the_stack (1000)        | 0✓            | 4.16e-237✓ | 1.16e-211✓    | 2.30e-76✓ | 0.999✓    | 0.313× | 0.995✓ | 0.976✓ |
| the_stack (123)         | 0.999✓        | 0.993✓     | 0.610✓        | 0.491✓    | 0.916×    | 0.255✓ | 0.328✓ | 0.038✓ |

$\alpha = 0.01$ . The absolute magnitude of extremely small  $p$ -values should not be overinterpreted: once  $p$  falls below numerical and sampling noise, e.g.,  $< 10^{-20}$ , values such as  $10^{-260}$  and  $10^{-20}$  both decisively reject the null rather than indicating meaningfully different lineage strength.

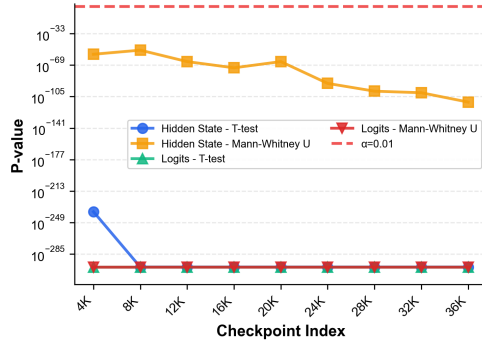
**Different initialization seeds produce distinct fingerprints** Table 9a and Table 9b report  $p$ -values from correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the  $t$ -test and  $U$ -test for LLaMA-style and Qwen-style models, respectively. All  $p$ -values are consistently  $> 0.01$ , indicating that the method separates models trained from different seeds. This supports seed-level identity across both model styles: distinct seeds yield distinct fingerprint behaviors at birth.

**Training preserves the initialization fingerprint.** Table 10 (LLaMA-style) and Table 11 (Qwen-style) compare each initialization model  $s^{init}$  with its descendant  $s^{base}$  after training on OpenWebText (Gokaslan et al., 2019) ( $\approx 10B$  tokens). Across all seed–model pairs,  $p$ -values are consistently  $< 0.01$ , showing that the bias profiles remain strongly correlated and indicate shared lineage. Thus, trained descendants retain the fingerprint of their initialization. The baseline methods in Table 10 fail to distinguish across seeds, suggesting that their separability comes from training-induced artifacts rather than initialization-level identity.

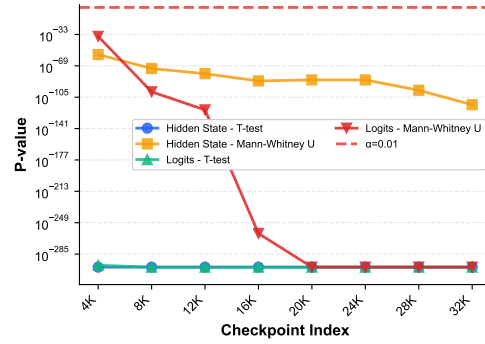
**Identical data and order do not make fingerprints converge** In Table 13a (LLaMA-style models) and Table 13b (Qwen-style models), all four “suspicious” models  $s_i^{base}$  for  $i \in \{42, 123, 100, 2000\}$  are trained on *exactly the same corpus (OpenWebText) and in the same data order* by fixing the training seed; only their initialization seeds  $i$  differ. This tests whether identical data and order erase or confound fingerprint behavior. Across all cross-seed pairs,  $p$ -values remain  $> 0.01$ , in contrast to

Table 13: Shared data and training order do not make fingerprint behavior identical across different initializations.

| (a) LLaMA-style models                  |                |                |                |                | (b) Qwen-style models                  |                |                |                |                |
|-----------------------------------------|----------------|----------------|----------------|----------------|----------------------------------------|----------------|----------------|----------------|----------------|
| Model Pair                              | Logits Output  |                | Hidden State   |                | Model Pair                             | Logits Output  |                | Hidden State   |                |
|                                         | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |                                        | <i>t</i> -test | <i>U</i> -test | <i>t</i> -test | <i>U</i> -test |
| $s_{123}^{init}$ vs. $s_{1000}^{base}$  | 0.484          | 0.500          | 0.385          | 0.486          | $s_{123}^{init}$ vs. $s_{1000}^{base}$ | 0.598          | 0.638          | 0.286          | 0.254          |
| $s_{1000}^{init}$ vs. $s_{2000}^{base}$ | 0.946          | 0.956          | 0.135          | 0.096          | $s_{1000}^{init}$ vs. $s_{123}^{base}$ | 0.804          | 0.805          | 0.236          | 0.240          |
| $s_{42}^{init}$ vs. $s_{123}^{base}$    | 0.598          | 0.589          | 0.426          | 0.337          | $s_{42}^{init}$ vs. $s_{2000}^{base}$  | 0.589          | 0.608          | 0.226          | 0.2043         |
| $s_{2000}^{init}$ vs. $s_{42}^{base}$   | 0.756          | 0.781          | 0.388          | 0.287          | $s_{2000}^{init}$ vs. $s_{42}^{base}$  | 0.523          | 0.482          | 0.312          | 0.323          |



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: Fingerprint verification remains significant at every checkpoint ( $p$ -values  $< 0.01$ ) across model families.

the near-zero values in Table 10. Fingerprints therefore remain seed-specific even under identical data and curriculum.

**Continual training on diverse datasets does not confound the fingerprint** The preceding experiments show that SeedPrints can function as a biometric-style fingerprint by tracking initialization identity. For copyright auditing, however, seed distinguishability is not the primary concern because initialization seeds have no clear copyright status. The broader fragility is that attribution methods can be misled by *data distribution, failing to recognize lineage when the training distribution shifts substantially during continued training.*

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base with (ii) *distractors* derived from a different base model, initialized with seed 123 and trained with a different OpenWebText data order, then continued on the *same* target corpus. This setup tests whether attribution methods recover true lineage rather than matching the downstream data domain.

Prior baselines all fail in the code setting (The Stack), misclassifying true descendants as distractors. This indicates that they largely track domain similarity rather than lineage identity: TinyStories is closer to OpenWebText, while The Stack differs sharply, and the larger distribution shift bypasses detection. SeedPrints correctly attributes lineage across both corpora, showing that the fingerprint is not merely a proxy for data distribution in these experiments and remains detectable beyond initial pretraining.

#### D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

**Controlled pretraining trajectory.** We conduct a controlled pretraining experiment on OpenWebText where the initialization seed is known. Each intermediate checkpoint is treated as the base, and we test whether the method identifies descendants along the same training trajectory. All variants recognize the suspect model as same-lineage, with  $p$ -values below the 0.01 threshold (Figure 5a for

Table 14: Cross-size lineage scores between Qwen-2.5-14B (target) and other Qwen-2.5 models.

|                          | Qwen-2.5-0.5B          | Qwen-2.5-1.5B          | Qwen-2.5-3B            | Qwen-2.5-7B            |
|--------------------------|------------------------|------------------------|------------------------|------------------------|
| SeedPrint ( $\uparrow$ ) | 0.7184                 | 0.9707                 | 0.3898                 | 0.5429                 |
| REEF ( $\downarrow$ )    | 0.8250                 | 0.8399                 | 0.8833                 | 0.8461                 |
| HUREF ( $\downarrow$ )   | $-6.37 \times 10^{-5}$ | $-2.84 \times 10^{-5}$ | $-8.70 \times 10^{-6}$ | $-6.08 \times 10^{-6}$ |

LLaMA-style models and Figure 5b for Qwen-style models). This supports SeedPrints stability under continuous optimization.

### D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To examine whether SeedPrints collapses to a family-level signature when parameter size changes, we report cross-size comparisons between Qwen-2.5-14B as the target and smaller models from the same Qwen-2.5 family. Table 14 reports SeedPrint ( $\uparrow$ ) and two baselines, REEF ( $\downarrow$ ) and HUREF ( $\downarrow$ ).

We therefore conduct a cross-size study within the Qwen-2.5 series. Qwen-2.5-14B is the protected target, while Qwen-2.5-{0.5B, 1.5B, 3B, 7B} are suspect models that share the same architectural family but are not trained as descendants of Qwen-2.5-14B. As discussed in the main paper, SeedPrints outputs a  $p$ -value for the hypothesis test that the suspect shares initialization-born fingerprints with the target, whereas REEF and HUREF output continuous similarity scores that require user-chosen thresholds.

Several observations are relevant for scope. First, all SeedPrints  $p$ -values are comfortably above a typical significance level such as  $\alpha=0.01$ . Under the hypothesis-testing view, none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This behavior is desirable: although these models belong to the same family and use similar architectural designs, the fingerprinting test does not collapse into a family-level signature that spuriously conflates distinct pretraining runs.

The cross-size  $p$ -values occupy different parts of the  $[0, 1]$  range rather than concentrating at one value. We do *not* interpret these magnitudes as continuous “similarity scores”; their role is to support a calibrated decision at a fixed significance level. Nevertheless, the spread suggests that the test statistic remains sensitive to the concrete combination of architecture and initialization seed rather than degenerating into a generic pattern shared by all Qwen-2.5 variants.

The similarity-based baselines behave differently. REEF produces scores in a narrow band (0.8250–0.8833) across sizes, while HUREF outputs values numerically close to zero (on the order of  $10^{-5}$ ). Because these methods lack a built-in decision rule, operators must choose thresholds manually, and a single cutoff may not robustly separate cross-size variants from genuinely lineage-related models. SeedPrints instead yields a statistical decision at a prescribed significance level, giving a clearer operational answer to whether two models plausibly share a training lineage within a model family.

## E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family. For each derived model we also indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” refer to *pre-trained* and *instruction-tuned* source models, respectively.

| Parameter        | Value |
|------------------|-------|
| max_new_tokens   | 512   |
| temperature      | 0.7   |
| top_p            | 0.9   |
| top_k            | 50    |
| do_sample        | true  |
| max_input_length | 512   |

## E.1 MODEL LIST BY FAMILY

### E.1.1 QWEN-2.5-7B

| Variant | HuggingFace repo                             | Type         |
|---------|----------------------------------------------|--------------|
| PT      | Qwen/Qwen2.5-7B                              | —            |
| IT      | Qwen/Qwen2.5-7B-Instruct                     | —            |
| Derived | Qwen/Qwen2.5-Math-7B                         | finetune     |
| Derived | Qwen/Qwen2.5-Coder-7B-Instruct               | finetune     |
| Derived | WangCa/Qwen2.5-7B-Medicine                   | finetune     |
| Derived | huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2 | finetune     |
| Derived | Locutusque/StockQwen-2.5-7B                  | merge        |
| Derived | bunnycore/QevaCoT-7B-Stock                   | merge        |
| Derived | fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct | adapter      |
| Derived | SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct    | adapter      |
| Derived | Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4           | quantization |
| Derived | Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8           | quantization |
| Derived | Lansechen/Qwen2.5-7B-Open-R1-Distill         | distillation |

### E.1.2 QWEN2.5-14B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT      | Qwen/Qwen2.5-14B                                | —            |
| IT      | Qwen/Qwen2.5-14B-Instruct                       | —            |
| Derived | Qwen/Qwen2.5-Coder-14B                          | finetune     |
| Derived | oxyapi/oxy-1-small                              | finetune     |
| Derived | v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno | merge        |
| Derived | ToastyPigeon/qwen-story-test-qlora              | adapter      |
| Derived | Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4             | quantization |
| Derived | deepseek-ai/DeepSeek-R1-Distill-Qwen-14B        | distillation |

### E.1.3 LLAMA-3.1-8B

| Variant | HuggingFace repo                                                 | Type         |
|---------|------------------------------------------------------------------|--------------|
| PT      | meta-llama/Llama-3.1-8B                                          | —            |
| IT      | meta-llama/Llama-3.1-8B-Instruct                                 | —            |
| Derived | ValiantLabs/Llama3.1-8B-Fireplace2                               | finetune     |
| Derived | RedHatAI/Llama-3.1-8B-tldr                                       | finetune     |
| Derived | proxectonos/Llama-3.1-Carballo                                   | finetune     |
| Derived | mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated                  | finetune     |
| Derived | gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES       | merge        |
| Derived | XiaoJian9992024/Llama3.1-8B-ExtraMix                             | merge        |
| Derived | LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching | adapter      |
| Derived | chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7                     | adapter      |
| Derived | iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8                   | quantization |
| Derived | DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ                            | quantization |
| Derived | asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill                    | distillation |

#### E.1.4 MISTRAL-7B-v0.3

| Variant | HuggingFace repo                                    | Type         |
|---------|-----------------------------------------------------|--------------|
| PT      | mistralai/Mistral-7B-v0.3                           | —            |
| IT      | mistralai/Mistral-7B-Instruct-v0.3                  | —            |
| Derived | KurmaAI/AQUA-7B                                     | finetune     |
| Derived | openfoodfacts/spellcheck-mistral-7b                 | finetune     |
| Derived | grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B      | merge        |
| Derived | chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer     | adapter      |
| Derived | RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit         | quantization |
| Derived | eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b | distillation |

#### E.1.5 GEMMA-2-2B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT      | google/gemma-2-2b                               | —            |
| IT      | google/gemma-2-2b-it                            | —            |
| Derived | rinna/gemma-2-baku-2b                           | finetune     |
| Derived | anakin87/gemma-2-2b-neogenesis-ita              | finetune     |
| Derived | vonjack/gemma2-2b-merged                        | merge        |
| Derived | google-cloud-partnership/gemma-2-2b-it-lora-sql | adapter      |
| Derived | qilowoq/gemma-2-2b-it-4Bit-GPTQ                 | quantization |
| Derived | Syed-Hasan-8503/Gemma-2-2b-it-distilled         | distillation |

#### E.1.6 LLAMA-2-7B

| Variant | HuggingFace repo                                                | Type         |
|---------|-----------------------------------------------------------------|--------------|
| PT      | meta-llama/Llama-2-7b-hf                                        | —            |
| IT      | meta-llama/Llama-2-7b-chat-hf                                   | —            |
| Derived | allenai/tulu-2-7b                                               | finetune     |
| Derived | QIAIUNCC/EYE-Llama_qa                                           | finetune     |
| Derived | DevQuasar/coma-7B-v0.1                                          | merge        |
| Derived | Ammar-1/llama2-Better-Tune                                      | adapter      |
| Derived | TheBloke/Llama-2-7B-Chat-GPTQ                                   | quantization |
| Derived | cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2 | distillation |

#### E.1.7 TINYLLAMA-1.1B

| Variant | HuggingFace repo                                | Type         |
|---------|-------------------------------------------------|--------------|
| PT/IT   | TinyLlama/TinyLlama-1.1B-Chat-v1.0              | —            |
| Derived | alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2 | finetune     |
| Derived | Edentns/DataVortexTL-1.1B-v0.1                  | finetune     |
| Derived | appvoid/dot-v2.7                                | merge        |
| Derived | barissglc/tinyllama-tarot-v1                    | adapter      |
| Derived | TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ          | quantization |
| Derived | anudaw/distilled-code-llama                     | distillation |

**Notes.** “Type” denotes the post-training transformation relative to the PT/IT source model: *finetune* covers supervised and preference-optimization variants; *adapter* covers LoRA/QLoRA-style modules; *merge* covers model soups and TIES-style merges; *quantization* covers GPTQ/INTx variants; and *distillation* covers student models distilled from reasoning-augmented teachers.

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## E.2 SAMPLE RESULTS

We present partial results from one LeafBench evaluation run in Table 15 ( $u$ -test with simulation-based null). The reported metric is  $1 - p$ , used only as a similarity-style score for benchmark compatibility. We use a 0.01 significance level for  $p$ -values; empirically, LeafBench identifies an optimal decision threshold of 0.9920, which is consistent with reliable separation in this run.

Importantly, individual  $p$ -values should not be read as linear similarity measures. Their distribution across model pairs is nevertheless informative: same-lineage models produce much lower  $p$ -values, while unrelated models remain closer to uniform.

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| Model                                                      | Qwen-2.5-7B | Qwen-2.5-7B-Instruct | Qwen2.5-14B | Qwen2.5-14B-Instruct |
|------------------------------------------------------------|-------------|----------------------|-------------|----------------------|
| Qwen-2.5-7B                                                | 1.000       | 1.000                | 0.079       | 0.000                |
| Qwen-2.5-7B-Instruct                                       | 1.000       | 1.000                | 0.813       | 0.536                |
| Qwen2.5-7B-Math                                            | 1.000       | 0.994                | 0.047       | 0.987                |
| Qwen2.5-7B-Coder                                           | 0.996       | 0.999                | 0.177       | 0.002                |
| Qwen2.5-7B-Instruct-Medicine                               | 1.000       | 1.000                | 0.576       | 0.369                |
| Qwen2.5-7B-Instruct-Abiliterated                           | 1.000       | 1.000                | 0.276       | 0.886                |
| Qwen2.5-7B-Stock                                           | 1.000       | 1.000                | 0.939       | 0.225                |
| QevaCoT-7B                                                 | 1.000       | 1.000                | 0.199       | 0.119                |
| Qwen2.5-7B-Instruct-Task-10                                | 1.000       | 1.000                | 0.672       | 0.090                |
| Qwen2.5-7B-Instruct-Task-12                                | 1.000       | 1.000                | 0.155       | 0.987                |
| Qwen2.5-7B-Instruct-Int4                                   | 1.000       | 1.000                | 0.827       | 0.961                |
| Qwen2.5-7B-Instruct-Int8                                   | 1.000       | 1.000                | 0.914       | 0.739                |
| Qwen2.5-7B-Open-R1-Distill                                 | 1.000       | 1.000                | 0.302       | 0.955                |
| Qwen2.5-14B                                                | 0.324       | 0.487                | 1.000       | 1.000                |
| Qwen2.5-14B-Instruct                                       | 0.403       | 0.702                | 1.000       | 1.000                |
| Qwen2.5-Coder-14B                                          | 0.492       | 0.660                | 1.000       | 1.000                |
| oxy-1-small                                                | 0.492       | 0.104                | 1.000       | 1.000                |
| Qwen2.5-14B-Gutenberg-Instruct-Slerpeno                    | 0.856       | 0.931                | 1.000       | 1.000                |
| Qwen-story-test-qlora                                      | 0.112       | 0.898                | 1.000       | 1.000                |
| Qwen2.5-14B-Instruct-GPTQ-Int4                             | 0.583       | 0.092                | 1.000       | 1.000                |
| DeepSeek-R1-Distill-Qwen-14B                               | 0.204       | 0.000                | 1.000       | 1.000                |
| Llama-3.1-8B                                               | 0.638       | 0.714                | 0.156       | 0.000                |
| Llama-3.1-8B-Instruct                                      | 0.643       | 0.118                | 0.128       | 0.510                |
| Llama-3.1-8B-Fireplace2                                    | 0.715       | 0.532                | 0.065       | 0.354                |
| Llama-3.1-8B-TLDR                                          | 0.925       | 0.811                | 0.331       | 0.014                |
| Llama-3.1-8B-Carballo                                      | 0.001       | 0.605                | 0.003       | 0.946                |
| Llama-3.1-8B-Instruct-Abliterated                          | 0.061       | 0.005                | 0.121       | 0.422                |
| Llama-3.1-8B-Instruct-HalfAbliterated-TIES                 | 0.356       | 0.509                | 0.796       | 0.000                |
| Llama-3.1-8B-ExtraMix                                      | 0.640       | 0.160                | 0.721       | 0.875                |
| Llama-3.1-8B-Instruct-cv-job-description-matching          | 0.684       | 0.292                | 0.004       | 0.001                |
| Llama-3.1-8B-Instruct-PsyCourse-fold7                      | 0.682       | 0.057                | 0.063       | 0.191                |
| Llama-3.1-8B-Instruct-8bit                                 | 0.798       | 0.214                | 0.154       | 0.385                |
| Llama-3.1-8B-Instruct-4bit                                 | 0.513       | 0.023                | 0.005       | 0.933                |
| Llama-3.1-8B-Instruct-Open-R1-Distill                      | 0.800       | 0.101                | 0.772       | 0.024                |
| Mistral-7B-v0.3                                            | 0.224       | 0.352                | 0.506       | 0.044                |
| Mistral-7B-v0.3-Instruct                                   | 0.000       | 0.000                | 0.984       | 0.104                |
| AQUA-7B                                                    | 0.014       | 0.613                | 0.703       | 0.569                |
| Mistral-7B-v0.3-Spellcheck                                 | 0.005       | 0.481                | 0.872       | 0.105                |
| Mistral-7B-v0.3-Instruct-demi-merge                        | 0.904       | 0.774                | 0.058       | 0.522                |
| Mistral-7B-v0.3-Brain                                      | 0.340       | 0.983                | 0.481       | 0.016                |
| Mistral-7B-v0.3-Instruct-GPTQ-4bit                         | 0.002       | 0.281                | 0.089       | 0.784                |
| Mistral-7B-distilled-from-deepseek-r1-qwen32b              | 0.009       | 0.001                | 0.511       | 0.015                |
| Gemma-2-2b                                                 | 0.747       | 0.869                | 0.689       | 0.000                |
| Gemma-2-2b-it                                              | 0.944       | 0.039                | 0.626       | 0.060                |
| Gemma-2-baku-2b                                            | 0.616       | 0.231                | 0.176       | 0.464                |
| Gemma-2-2b-neogenesis-ita                                  | 0.002       | 0.536                | 0.079       | 0.980                |
| Gemma-2-2b-merged                                          | 0.951       | 0.012                | 0.651       | 0.075                |
| Gemma-2-2b-it-lora-sql                                     | 0.236       | 0.408                | 0.000       | 0.000                |
| Gemma-2-2B-it-4Bit-GPTQ                                    | 0.945       | 0.159                | 0.000       | 0.008                |
| Gemma-2-2b-it-distilled                                    | 0.482       | 0.573                | 0.271       | 0.276                |
| Llama-2-7b                                                 | 0.443       | 0.470                | 0.148       | 0.276                |
| Llama-2-7b-chat                                            | 0.528       | 0.868                | 0.424       | 0.766                |
| tulu-2-7b                                                  | 0.125       | 0.272                | 0.170       | 0.888                |
| EYE-Llama_qa                                               | 0.657       | 0.244                | 0.674       | 0.968                |
| coma-7B-v0.1                                               | 0.953       | 0.004                | 0.573       | 0.731                |
| llama2-Better-Tune                                         | 0.136       | 0.699                | 0.469       | 0.860                |
| Llama-2-7B-Chat-GPTQ                                       | 0.594       | 0.603                | 0.582       | 0.424                |
| llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2 | 0.349       | 0.224                | 0.366       | 0.976                |

Table 15: **Sample results.** Pairwise lineage scores between target models (rows) and base models (columns). Cells with  $1 - p \geq 0.99$  are highlighted in green as same-lineage pairs.